LIFTING AN ALGEBRA MAP B TO C MODULO A DEEPER POWER OF I

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof computes the Ext identifications, applies the uniform bound to push the obstruction past I^{n'}, observes 2n' > n from the hypothesis on n, iterates the lifting, and takes the limit in the complete ring C

Editorial note. The label lemma-get-morphism-general-better names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the obstruction class o(psi_n) in Ext^1_B(NL^wedge_{B/A}, I^nC/I^{2n}C) attached to a map psi_n : B -> C/I^nC, constructed in the same chapter
- the adjunction identification of Ext after tensoring with C/I^nC and the two-term base change identification, both from More on Algebra
- the uniform bound lemma of Local Cohomology: integers d and q_0 of the associated graded Gr_I(C) control the annihilation of Ext classes of two-term complexes
- I-adic completeness of C and taking limits of compatible algebra maps

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at proof line(s) 1172; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GAD: sources/stacks-chow/tags.txt, source lines 20333–20333. Prerequisite pointer invoked at proof line(s) 1207; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0GA6: sources/stacks-chow/tags.txt, source lines 20326–20326. Prerequisite pointer invoked at proof line(s) 1177; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0G9G: sources/stacks-chow/tags.txt, source lines 20301–20301. Prerequisite pointer invoked at proof line(s) 1202; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0E1W: sources/stacks-chow/tags.txt, source lines 17586–17586. Prerequisite pointer invoked at proof line(s) 1200; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GAP: sources/stacks-restricted/pinned-original-source.tex, source lines 1101–1101. Prerequisite pointer invoked at proof line(s) 1190; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Assume given the following data*

(1) *an integer $c \geq 0$,*
(2) *an ideal I of a Noetherian ring A,*
(3) *B in (0AL4) for (A, I) such that I^c annihilates $\mathrm{Ext}^1_B(NL^\wedge_{B/A}, N)$ for any B-module N,*
(4) *a Noetherian I-adically complete A-algebra C; denote $d = d(Gr_I(C))$ and $q_0 = q(Gr_I(C))$ the integers found in Local Cohomology, Section 0GA6,*
(5) *an integer $n \geq \max(q_0 + (d+1)c, 2(d+1)c + 1)$, and*
(6) *an A-algebra homomorphism $\psi_n : B \to C/I^nC$.*

Then there exists a map $\varphi : B \to C$ of A-algebras such that $\psi_n \bmod I^{n-(d+1)c} = \varphi \bmod I^{n-(d+1)c}$.

Proof. Consider the obstruction class

$$o(\psi_n) \in \mathrm{Ext}^1_B(NL^\wedge_{B/A}, I^nC/I^{2n}C)$$

of Remark 0GAP. For any C/I^nC-module N we have

$$\begin{aligned}\mathrm{Ext}^1_B(NL^\wedge_{B/A}, N) &= \mathrm{Ext}^1_{C/I^nC}(NL^\wedge_{B/A} \otimes^{\mathbf{L}}_B C/I^nC, N) \\ &= \mathrm{Ext}^1_{C/I^nC}(NL^\wedge_{B/A} \otimes_B C/I^nC, N)\end{aligned}$$

The first equality by More on Algebra, Lemma 0E1W and the second one by More on Algebra, Lemma 0G9G. In particular, we see that $\mathrm{Ext}^1_{C/I^nC}(NL^\wedge_{B/A} \otimes_B C/I^nC, N)$ is annihilated by I^cC for all C/I^nC-modules N. It follows that we may apply Local Cohomology, Lemma 0GAD to see that $o(\psi_n)$ maps to zero in

$$\mathrm{Ext}^1_{C/I^nC}(NL^\wedge_{B/A} \otimes_B C/I^nC, I^{n'}C/I^{2n'}C) = \mathrm{Ext}^1_B(NL^\wedge_{B/A}, I^{n'}C/I^{2n'}C) =$$

where $n' = n - (d+1)c$. By the discussion in Remark 0GAP we obtain a map

$$\psi'_{2n'} : B \to C/I^{2n'}C$$

which agrees with ψ_n modulo $I^{n'}$. Observe that $2n' > n$ because $n \geq 2(d+1)c+1$.

We may repeat this procedure. Starting with $n_0 = n$ and $\psi^0 = \psi_n$ we end up getting a strictly increasing sequence of integers

$$n_0 < n_1 < n_2 < \ldots$$

and A-algebra homorphisms $\psi^i : B \to C/I^{n_i}C$ such that ψ^{i+1} and ψ^i agree modulo $I^{n_i - tc}$. Since C is I-adically complete we can take φ to be the limit of the maps $\psi^i \bmod I^{n_i-(d+1)c} : B \to C/I^{n_i-(d+1)c}C$ and the lemma follows. □